

APPLIANCE ENERGY FORECASTING

Time-Series Modelling and 24-Hour Forecasting

GITHUB LINK: https://github.com/IswaryaLaharikola/Time_Series

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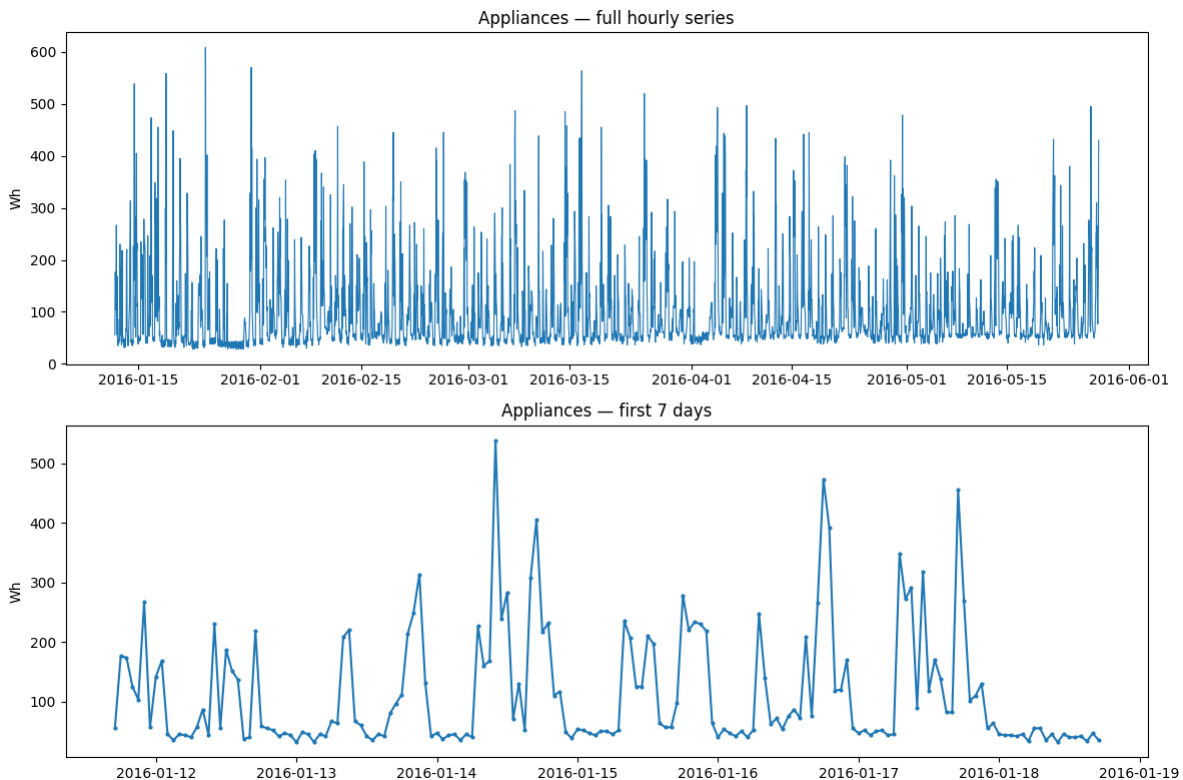
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1. Introduction

Accurate energy forecasting is becoming increasingly important for smart-home energy management and efficient electricity use. Household appliance consumption changes throughout the day, showing both regular patterns and sudden peaks. This assignment analyses historical appliance energy data and develops models to forecast the next 24 hours of energy consumption. Several approaches, including benchmark methods, SARIMAX, Random Forest and Chronos, are compared using standard forecasting metrics. The aim is to determine which approach provides the most accurate, reliable and practical forecast for household appliance energy demand.

2. Dataset and Preprocessing

The UCI Appliances Energy Prediction **dataset** contains 19,735 observations and 28 variables, recorded at 10-minute intervals. The data include appliance energy consumption and environmental measurements. The timestamps were converted to datetime format and checked for missing values and time gaps, with no missing data identified. The observations were then aggregated into hourly averages, resulting in 3,290 observations. The rv1 and rv2 random variables were removed before the forecasting analysis.



3. Exploratory Time-Series Analysis

3.1. Stationarity

The Augmented Dickey-Fuller (ADF) test was used to check whether the hourly appliance energy series was stationary. The original series produced an ADF statistic of **-8.949** with a **p-value below 0.001**, indicating that the series is stationary. The first-differenced series also remained stationary, with an ADF statistic of **-16.953** and a p-value below 0.001. These results support the use of time-series forecasting methods without requiring extensive differencing.

3.2. ACF and PACF

The ACF and PACF plots were used to examine the autocorrelation and dependence structure of the hourly appliance energy series. The ACF shows noticeable patterns around lag 24, indicating a strong daily seasonal cycle. Since the data are recorded hourly, 24 observations represent one day. This finding supports the use of a 24-hour seasonal period in the SARIMAX model.

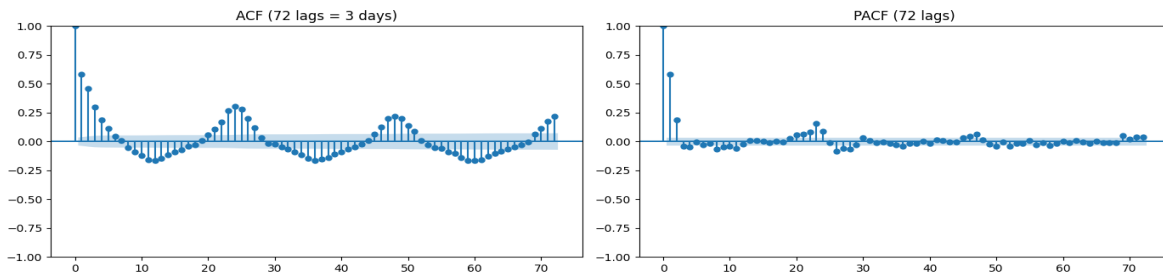


Figure 2. ACF and PACF of hourly appliance energy consumption.

3.3. Seasonal Decomposition

Seasonal decomposition was used to separate the appliance energy series into trend, seasonal and residual components. The results show a clear daily seasonal pattern, supporting the use of a 24-hour seasonal period. The residual component contains irregular fluctuations and consumption spikes that are more difficult to predict. These findings helped justify the use of seasonal time-series modelling in the forecasting analysis.

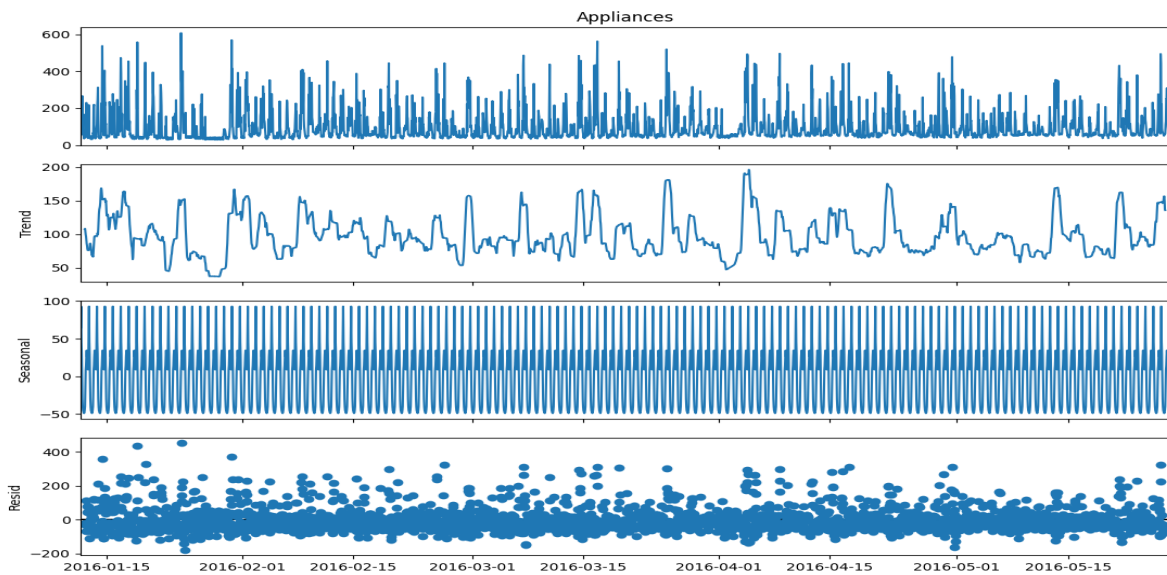


Figure 3. Seasonal decomposition using a 24-hour seasonal period.

4. Forecasting Methodology

The forecasting task was designed to predict the next 24 hours of appliance energy consumption using historical hourly data. A chronological train-test split was used to prevent future information from influencing the model. Several benchmark models were first established, followed by SARIMAX, Random Forest and Chronos. Model performance was compared using RMSE, MAE, MAPE and MASE. This consistent evaluation framework allowed the forecasting approaches to be compared fairly on the same test period.

Dataset Split	Start Date & Time	End Date & Time	Duration
Training	2016-01-11 17:00:00	2016-05-13 18:00:00	2,954 h
Test	2016-05-13 19:00:00	2016-05-14 18:00:00	24 h
Post-Test / Held-Out	2016-05-14 19:00:00	2016-05-27 18:00:00	312 h

5. Benchmark Models

Five benchmark models were tested: Mean, Naive, Daily Seasonal Naive, Weekly Seasonal Naive and Drift. These provided baseline results for comparison with the more advanced models. Weekly Seasonal Naive performed best, achieving an RMSE of 48.81, indicating that weekly patterns are useful for predicting appliance energy consumption.

Model	RMSE	MAE	MAPE %	MASE
Mean	50.01	39.38	40.46	0.74
Naive	247.60	242.64	298.67	4.54
Daily seasonal naive	118.04	80.28	70.61	1.50
Weekly seasonal naive	48.81	30.00	22.44	0.56
Drift	248.76	243.88	299.91	4.57

6. SARIMAX Modelling

SARIMAX was used to model the 24-hour seasonal pattern identified in the data. An AIC search tested 147 combinations of p,d,q, resulting in SARIMAX(4,1,6) × (1,0,1,24). Residual diagnostics were used to assess model fit, and a 24-hour forecast with 95% confidence intervals was generated. The model achieved an RMSE of 32.37, outperforming the benchmark models. This demonstrates that incorporating both temporal dependence and daily seasonality improved the forecasting accuracy.

Model component	Result
AIC combinations	147
Selected order	(4,1,6)
Seasonal order	(1,0,1,24)
Test RMSE	32.37
Test MAE	24.15

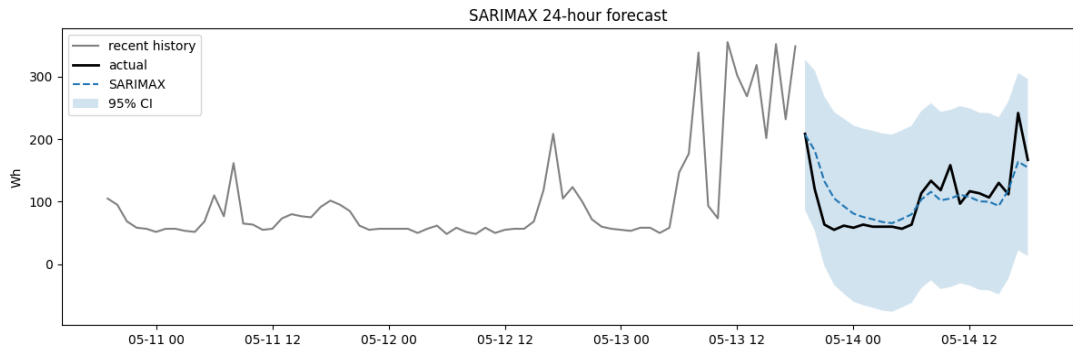


Figure 4. SARIMAX 24-hour forecast with 95% confidence interval.

6.1. Residual Diagnostics

The SARIMAX residuals were analysed using ACF and distribution plots to assess model fit. The residuals showed some irregular variation, with skewness of 2.03 and kurtosis of 11.60, indicating a right-skewed and heavy-tailed distribution. These results suggest that the model captures the main patterns but may not fully explain the sudden appliance-energy spikes.

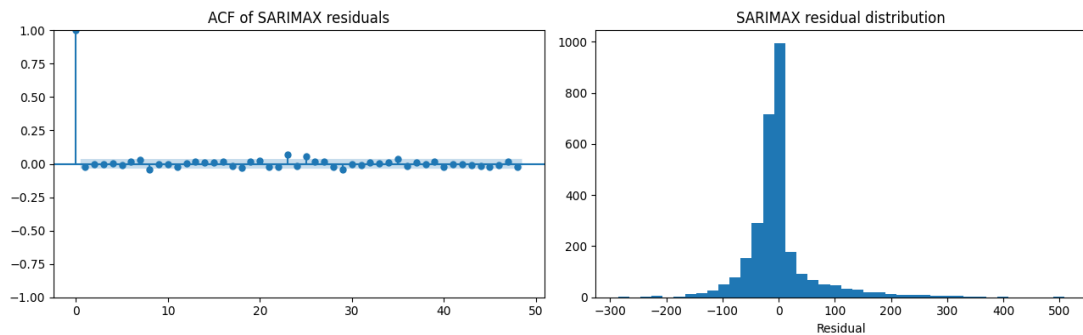


Figure 5. SARIMAX residual diagnostics.

Diagnostic	Result
Ljung-Box lag 1 p-value	0.99

Jarque-Bera	11010.02
Jarque-Bera p-value	<0.001
Skewness	2.03
Kurtosis	11.60

7. Feature-Based Random Forest

A Random Forest model was developed using time-based, lagged, rolling-window and environmental features to improve forecasting. The model used 62 features, with hour and historical appliance-consumption lags among the most important variables. It achieved an RMSE of 73.64, which was higher than SARIMAX and the weekly seasonal benchmark. This indicates that adding more features did not necessarily improve forecasting accuracy for this dataset.

Result	Value
Features	62
Trees	500
Max depth	5
Minimum leaf size	8
Best CV RMSE	68.38
Test RMSE	73.64
Test MAE	56.93

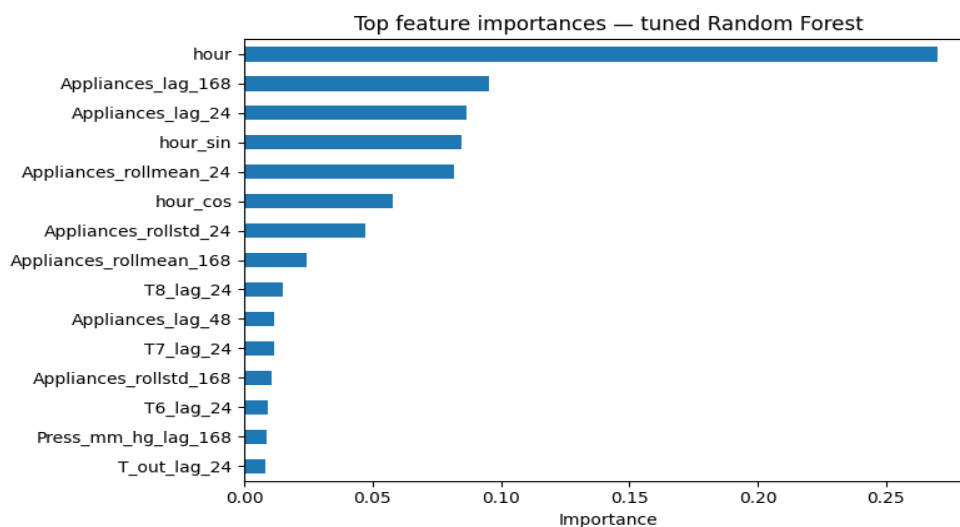


Figure 6. Random Forest feature importance.

8. Chronos Foundation Model

The Chronos foundation model was used to generate a 24-hour appliance energy forecast. It achieved an RMSE of 47.69, performing better than several simple benchmarks but worse than SARIMAX. This shows that Chronos provided a competitive forecast, but it did not improve the overall accuracy achieved by the statistical model.

Metric	Chronos
RMSE	47.69
MAE	33.36
MAPE	24.73%
MASE	0.62

9. Overall Model Comparison

The results show that SARIMAX performed best, achieving the lowest RMSE of 32.37. Chronos and Weekly Seasonal Naive followed with RMSE values of 47.69 and 48.81, respectively. Random Forest achieved 73.64, while the Naive and Drift models performed worst. Overall, SARIMAX provided the most accurate forecast by effectively capturing the temporal dependence and daily seasonal pattern in appliance energy consumption.

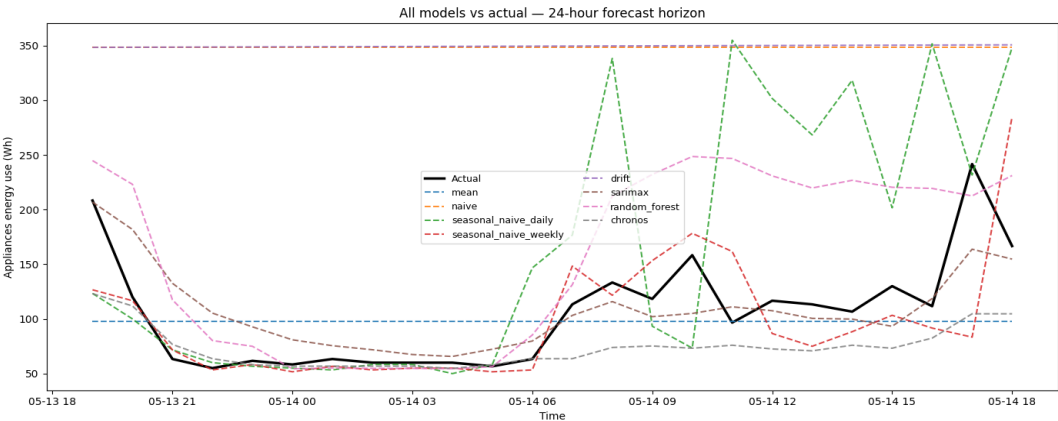


Figure 7. Forecast comparison across all models and actual observations.

Rank	Model	RMSE	MAE	MAPE %	MASE
1	SARIMAX	32.37	24.15	26.68	0.45
2	Chronos	47.69	33.36	24.73	0.62
3	Weekly seasonal naive	48.81	30.00	22.44	0.56
4	Mean	50.01	39.38	40.46	0.74
5	Random Forest	73.64	56.93	51.78	1.07
6	Daily seasonal naive	118.04	80.28	70.61	1.50
7	Naive	247.60	242.64	298.67	4.54

8	Drift	248.76	243.88	299.91	4.57
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10. Part 9 – Required Questions

1. Which benchmark model is strongest — naive, daily seasonal naive, weekly seasonal naive, or drift — and what does this tell you about the structure of appliance energy use?

Answer : The Weekly Seasonal Naive model was the strongest benchmark, achieving an RMSE of 48.81. It performed better than the Mean (50.01), Daily Seasonal Naive (118.04), Naive (247.60) and Drift (248.76) models. This suggests that weekly seasonal patterns and household routines provide useful information for predicting appliance energy consumption.

2. Does the SARIMAX model improve on the strongest seasonal benchmark? Discuss whether daily seasonality, autocorrelation, and exogenous variables are adequately captured.

Answer : Yes. SARIMAX achieved an RMSE of 32.37, compared with 48.81 for the Weekly Seasonal Naive model. The 24-hour seasonal component captured the strong daily pattern identified by the ACF and PACF analysis. The model also captured temporal dependence through its autoregressive and moving-average terms. However, the residual distribution remained right-skewed and heavy-tailed, suggesting that some appliance-consumption spikes were not fully captured.

3. Does the XGBoost or feature-based model improve when lag, rolling-window, time-of-day, and sensor/weather variables are added? Which feature groups appear most useful?

Answer : No. The Random Forest model achieved an RMSE of 73.64, which was worse than both SARIMAX and the Weekly Seasonal Naive model. The most useful features were hour of day (0.27), 168-hour appliance lag (0.10), 24-hour appliance lag (0.09), and rolling appliance statistics. This suggests that time-of-day and historical appliance consumption were more useful than the added sensor and weather variables.

4. Does the foundation model outperform the simpler benchmark, SARIMAX, and feature-based models? Is the improvement, if any, large enough to justify the extra complexity?

Answer : No. The Chronos foundation model achieved an RMSE of 47.69, which was better than Random Forest (73.64) but slightly worse than the Weekly Seasonal Naive model (48.81) and clearly worse than SARIMAX (32.37). Therefore, Chronos did not provide enough improvement to justify its additional complexity for this forecasting task.

5. Which variables would genuinely be known at the forecast origin? If you use future indoor temperature, humidity, or weather values from the test set, is this a true forecast or a conditional forecast?

Answer : At the forecast origin, time variables and historical appliance data are known. Future weather, temperature and sensor values are not normally available. Using actual future values introduces information from the future. Therefore, it becomes a conditional forecast rather than a true forecast.

6. Based on accuracy, interpretability, uncertainty, computational cost, and ease of deployment, which model would you recommend for practical smart-home energy forecasting, and why?

Answer : I would recommend SARIMAX for practical smart-home energy forecasting. It achieved the best accuracy with an RMSE of 32.37 and MAE of 24.15. It is also interpretable and provides 95% confidence intervals for forecasts. Although Chronos is more complex, it did not improve accuracy enough to justify the additional computational cost.

11. Comparison with Subsequent Observations

The later 312 observations were compared with the forecast period to assess future energy behaviour. The data continued to show daily patterns and occasional high-consumption spikes. These spikes were difficult for the

models to predict accurately. Overall, the results highlight the importance of capturing both seasonality and sudden energy changes.

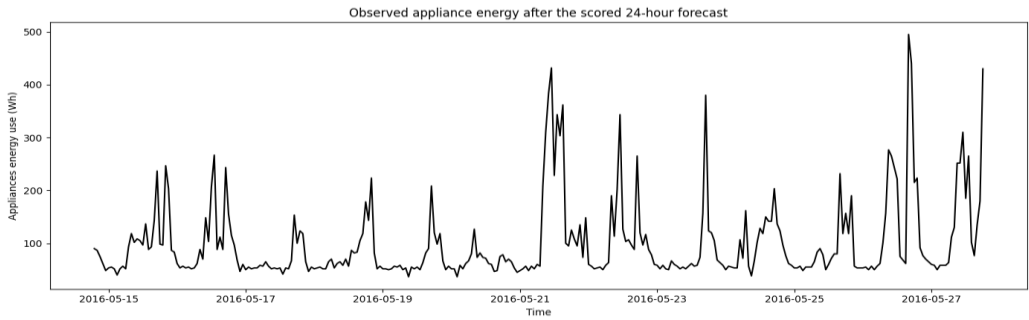


Figure 8. Observed appliance energy after the formal 24-hour test window.

12. Appendix A. Individual Model Forecasts

This appendix shows the 24-hour forecasts from all models. The plots compare predicted and actual appliance energy consumption. They provide additional evidence for the model comparison.

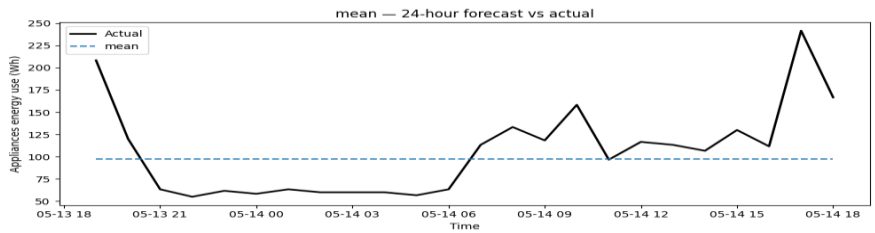


Figure A1. Mean forecast versus actual.

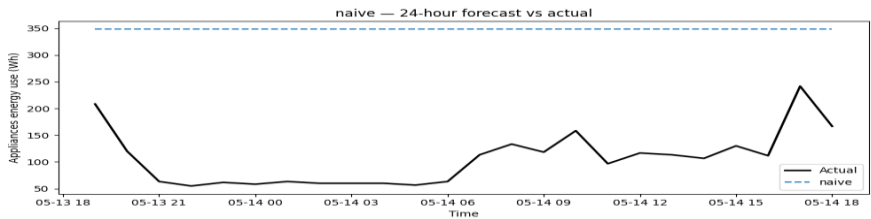


Figure A2. Naive forecast versus actual.

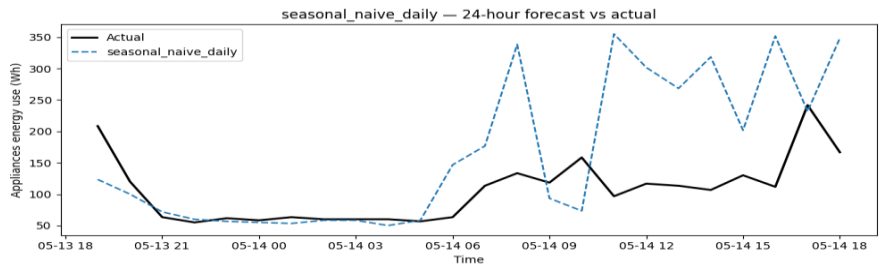


Figure A3. Seasonal Naive Daily forecast versus actual.

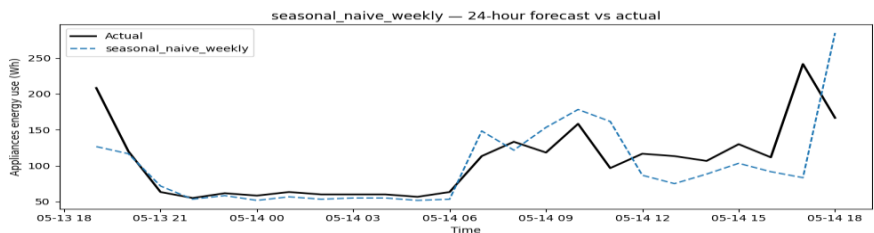


Figure A4. Seasonal Naive Weekly forecast versus actual.

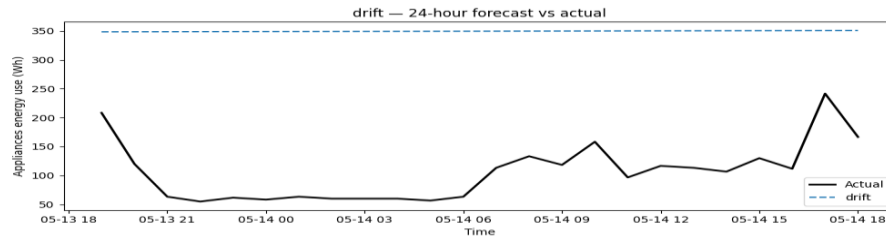


Figure A5. Drift forecast versus actual.

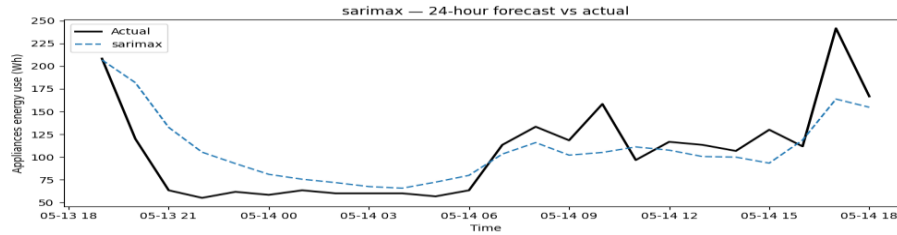


Figure A6. Sarimax forecast versus actual.

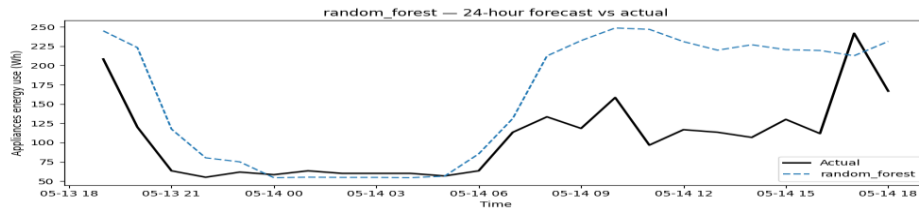


Figure A7. Random Forest forecast versus actual.

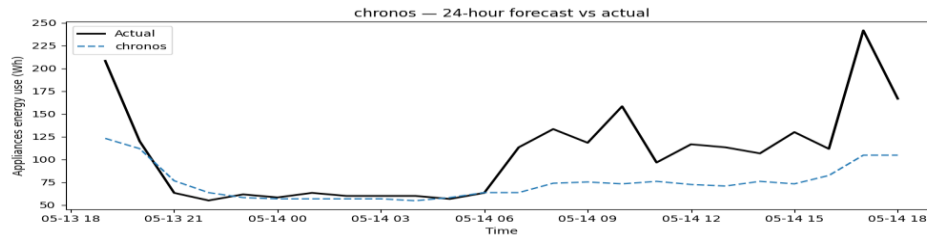


Figure A8. Chronos forecast versus actual.

13. Limitations and Future Improvements

- Only one 24-hour period was used for formal testing.
- Sudden energy-consumption spikes were difficult to predict.
- SARIMAX residuals showed skewness and heavy tails.
- Future weather and sensor values may not be available at forecast time.
- Future work could use multiple forecast periods for stronger evaluation.
- XGBoost or LightGBM could be tested with improved features..

14. Conclusion

The analysis identified clear daily and weekly patterns in household appliance energy consumption. Among the benchmark models, Weekly Seasonal Naive performed best, achieving an RMSE of 48.81. The SARIMAX model performed best overall, with an RMSE of 32.37 and MAE of 24.15. Random Forest and Chronos produced higher errors, showing that increased model complexity did not necessarily improve forecasting

accuracy. Overall, SARIMAX is recommended as the most suitable model for accurate and practical 24-hour appliance energy forecasting.

15. References

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